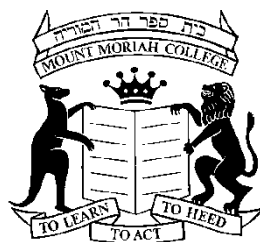


Student Number_____



MORIAH COLLEGE

Year 12 – Task 2 - Pre-Trial

MATHEMATICS 2014

Time Allowed: 3 hours

Examiners: G. Busuttil, O. Golan,

OUTCOMES ADDRESSED: P3,P5,H2,H4,H5,H6,H7,H8

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the end of this paper
- All necessary calculations should be shown in every question.

Section I

Multiple choice questions 1-10
10 marks

Section II

Short response questions 11-16
90 marks

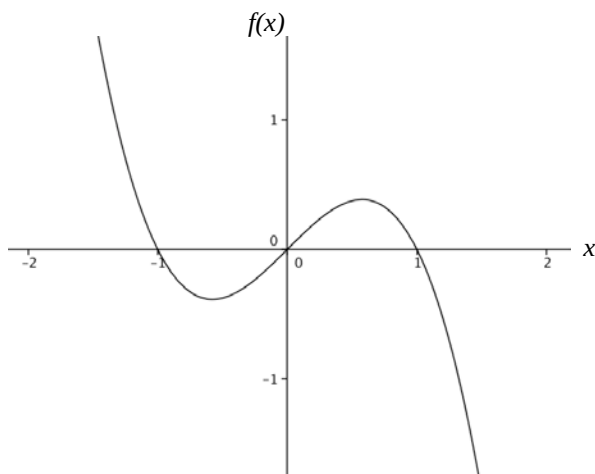
Total marks: 100

Section I Multiple Choice Questions. 1 mark each.

Circle the correct response on the answer sheet provided at the end of the paper.

- (1) Factorise $2h^2 - 11h + 15$
- A. $(2h+5)(h+3)$ B. $(2h+5)(h-3)$ C. $(2h-5)(h+3)$ D. $(2h-5)(h-3)$
- (2) How many terms are in the sequence $\sum_{n=7}^{30} 3n$
- A. 21 B. 22 C. 23 D. 24
- (3) The derivative of $y = \frac{4}{x^3}$ is
- A. $y' = \frac{-12}{x^4}$ B. $y' = \frac{-12}{x^{-4}}$ C. $y' = 4x^{-3}$ D. $y' = \frac{-2}{x^2}$
- (4) Solve $|2x-1| \leq 5$
- A. $-2 \geq x \geq 3$ B. $-2 \leq x \leq 3$ C. $x \leq -2, x \geq 3$ D. $x \leq -3, x \geq 2$
- (5) If $\log_t z = p$, then
- A. $z = p^t$ B. $p = z^t$ C. $z = t^p$ D. $p = t^z$
- (6) Factorise $a^3 - 64$
- A. $(a-4)(a^2 - 8a + 16)$ B. $(a-4)(a^2 + 8a + 16)$
- C. $(a-4)(a^2 - 4a + 16)$ D. $(a-4)(a^2 + 4a + 16)$

Questions 7 and 8 both refer to the function $f(x) = -x^3 + x$



(7)

$f(x)$ is:

- A. even
- B. odd
- C. neither
- D. unable to be determined

(8)

The integral of $f(x) = -x^3 + x$ from $x = -1$ to $x = 1$ is:

A. $2 \int_{-1}^0 (-x^3 + x) dx$ B. $2 \int_0^1 (-x^3 + x) dx$

- C. Either A or B D. 0

(9)

The exact value of $\sin 225^\circ$ is:

A. $\frac{\sqrt{3}}{2}$ B. $-\frac{\sqrt{3}}{2}$ C. $\frac{\sqrt{2}}{2}$ D. $-\frac{\sqrt{2}}{2}$

(10)

If $\log_c 2 = 0.46$, $\log_c 3 = 0.67$, $\log_c 5 = 1.27$, then $\log_c 30 = ?$

- A. 0.391414 B. 2.4 C. 1.1591 D. 6.7

Section II Short response Questions. 15 marks each.

Question 11 (START A NEW BOOKLET)

- (a) Calculate the perpendicular distance of the point (3, -1) from the line $4y = 3x + 2$. 2
- (b) Express $\frac{\log_3 8}{\log_3 2}$ as an integer 2
- (c) Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ 2
- (d) Determine the value of n to make the following expression equal to a **single** digit number: 2
- $$5^2 \times 2^4 \times 10^{-n}$$
- (e) Find the equation of the tangent to the curve $y = 5\log_e x$ at $x = 1$. 3
- (f) Solve for x : $(4x - 3)^2 = 25$ 2
- (g) If $(3 + \sqrt{3})^2 = a + b\sqrt{3}$, find the values of a and b . 2

END OF QUESTION 11

Question 12 (START A NEW BOOKLET)

(a) Differentiate with respect to x :

- | | | |
|------|--------------------|---|
| (i) | $x^2 e^{-x}$ | 2 |
| (ii) | $\frac{x^2}{3x+1}$ | 2 |

(b) Find $\int \frac{2x^2}{2x^3-3} dx$ 2

- (c) If α and β are the roots of the equation $3x^2 - 4x - 1 = 0$, find:
- | | | |
|------|---------------------------------|---|
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\alpha\beta$ | 1 |
| (ii) | $\alpha^{-1} + \beta^{-1}$ | 1 |
| (iv) | $\alpha\beta^3 + \beta\alpha^3$ | 2 |

- (d) (i) Solve the equation $5^{3x} = 0.04$ 2
- (ii) Solve $\log_2 x - \log_2 (x-3) = 2$ 2

END OF QUESTION 12

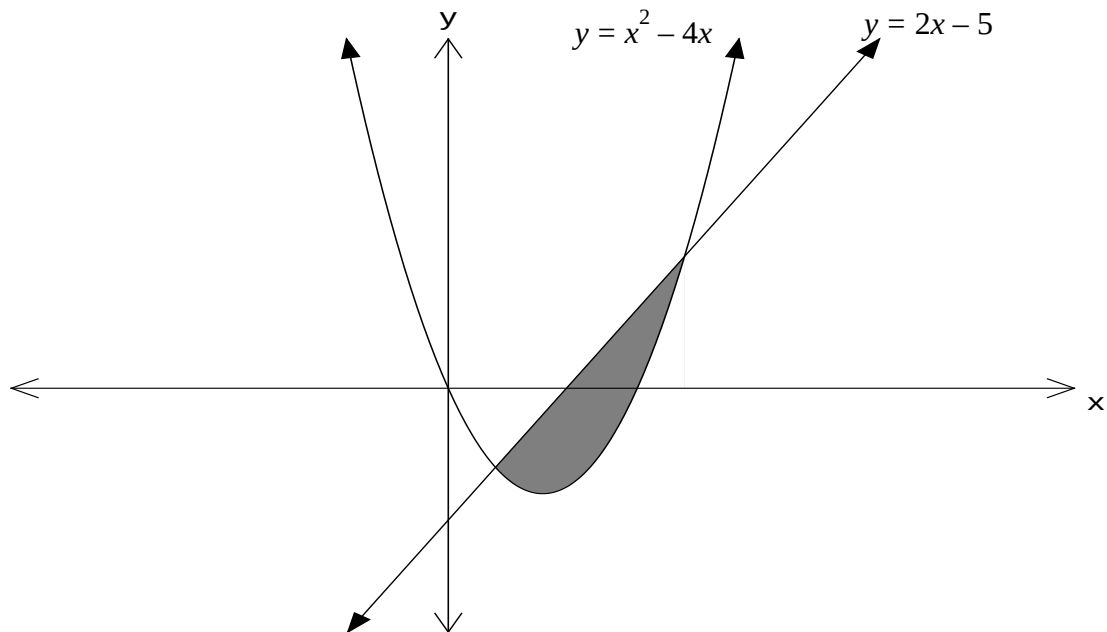
Question 13 (START A NEW BOOKLET)

- (a) The third term of an arithmetic progression is 23 and the tenth term is 72. 2
(i) Find the first term and the common difference. 2
(ii) Calculate the sum of the first 18 terms.
- (b) The first term of a geometric progression is 6 and the common ratio is 3. 2
How many terms of this progression are required to give a sum of 1594320?
- (c) The derivative of a function is given by $f'(x) = 15(5x - 1)^2$. 2
If $f(0) = 10$, find the equation $f(x)$
- (d) (i) Find the equation of the locus of $P(x, y)$, if P is always equidistant from $A(3, 1)$ and $B(1, 3)$. 2
(ii) Give a geometric description of this locus. 1
- (e) (i) On the same set of axes, graph $y = |2x - 1|$ and $y = -x$. 3
(ii) Use your graph, or otherwise, to explain why $|2x - 1| + x = 0$ has no solutions. 1

END OF QUESTION 13

Question 14 (START A NEW BOOKLET)

- (a) The graph shows the curves $y = x^2 - 4x$ and $y = 2x - 5$



- (i) Show the curves intersect when $x = 1$ and $x = 5$. 2
- (ii) Find the shaded area between the two curves 3
- (b) Solve $\log_7 x^2 = 3$.
Give your answer in exact simplified form. 2
- (c) Consider the function $f(x) = x^3 + 6x^2 + 9x + 4$ in the domain $-4 \leq x \leq 1$
- (i) Find the coordinates of any stationary points and determine their nature. 3
- (ii) Determine the coordinates of its point(s) of inflexion. 2
- (iii) Draw a sketch of the curve $y = f(x)$ in the domain $-4 \leq x \leq 1$ clearly showing all its essential features. 2
- (iv) What is the global maximum value of the function $y = f(x)$ in the domain $-4 \leq x \leq 1$? 1

END OF QUESTION 14

Question 15 (START A NEW BOOKLET)

- (a) Tom sets a pendulum swinging and notices that each swing is 80% as long as the preceding swing. The first swing is 20cm, the second swing is 16cm, and it continues to swing until coming to rest. 2

What is the total distance the pendulum swings?

- (b) Find the focus and directrix of the parabola $x^2 - 8x - 16y + 48 = 0$ 2

- (c) Prove that $\frac{\tan \theta \sec \theta}{1 + \tan^2 \theta} = \sin \theta$. 2

- (d) (i) Differentiate $y = xe^x$ 1

- (ii) Hence, evaluate $\int_0^2 \frac{xe^x}{2} dx$ 3

- (e) (i) Use the trapezoidal rule with 5 function values to find an approximation to 3

$$\int_0^2 \frac{1}{x+1} dx$$

- (ii) Calculate the difference between your answer in part (i) to the exact value, correct to 3 decimal places. 2

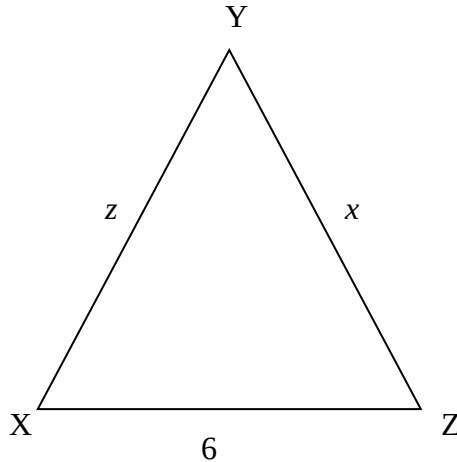
END OF QUESTION 15

Question 16 (START A NEW BOOKLET)

- (a) Find the maximum value of the function $y = -16x^2 + 160x - 256$ 2

- (b) Triangle XYZ has $XZ = 6$, $YZ = x$ and $XY = z$, as shown.

The perimeter of $\triangle XYZ$ is 16. All measurements are in centimetres.



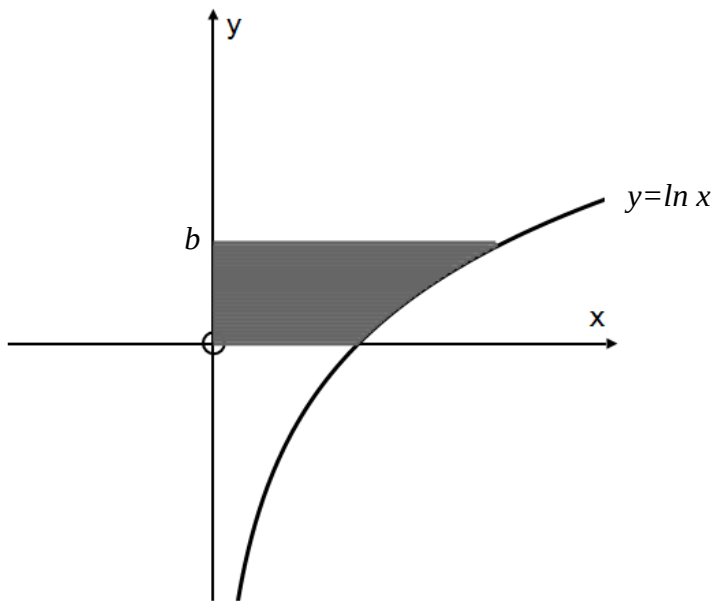
- (i) Express z in terms of x 1
- (ii) Using the cosine rule, express z^2 in terms of x and $\cos Z$ 1
- (iii) Hence, show that $\cos Z = \frac{5x - 16}{3x}$ 2
- (iv) Let the area of $\triangle XYZ$ be A . 1
Show $A^2 = 9x^2 \sin^2 Z$
- (v) Hence, show that $A^2 = -16x^2 + 160x - 256$ 2
- (vi) Using your answer from question (a), or otherwise, find the maximum area for $\triangle XYZ$? 1

Question 16 continues on the next page

Question 16 (Continued)

- (b) The quadratic equation $(k+1)x^2 - 4kx + 4k - 3 = 0$ has a root equal to 1.
Find k . 2

- (c) The shaded area is one square unit. Find the exact value of b . 3



END OF EXAM

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Student Number:

Teacher:

CIRCLE EACH CORRECT ANSWER.

MULTIPLE CHOICE ANSWER SHEET

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D

PRETRIAL MATHEMATICS

2014

SECTION 1 MULTIPLE CHOICE

1. $2h^2 - 11h + 15$
 $(2h-5)(h-3) \dots (D)$

1 D

2 D

2. $30 - 7 + 1 = 24 \dots (D)$

3 A

4 B

3. $y = 4x^{-3}$
 $y' = -12x^{-4}$
 $= \frac{-12}{x^4} \dots (A)$

5 C

6 D

7 B

8 D

9 D

4. $|2x-1| \leq 5$

10 B

$-5 \leq 2x-1 \leq 5$

$-4 \leq 2x \leq 6$

$-2 \leq x \leq 3 \dots (B)$

5. $\log_t z = p$ then
 $z = t^p \dots (C)$

6. $a^3 - 64$
 $= (a-4)(a^2 + 4a + 16) \dots (D)$

7. (B) odd

8. (D)

9. $\sin 225 \dots (B)$

10. $\log_c 30 = \log_c (2 \times 3 \times 5) = 0.46 + 0.67 + 1.27 = 2.4$

SECTION II

Question 11

$$a) d = \frac{3 \times 3 - 4(-1) + 2}{\sqrt{3^2 + 4^2}} \quad (1)$$

$$= \frac{9 + 4 + 2}{5}$$

$$= \frac{15}{5}$$

$$= \boxed{3} \quad (2)$$

$$b) \frac{\log_2 8}{\log_2 2} = \frac{\log_2 2^3}{\log_2 2} \quad (1)$$

$$= \boxed{3} \quad (2)$$

$$c) \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{(x-4)} \quad (1)$$

$$= \boxed{8} \quad (2)$$

$$d) 5^2 \times 2^4 \times 10^{-n} = \frac{25 \times 16}{10^n}$$

$$= \frac{400}{10^n} \quad (1) \text{ or } 10^n = 100$$

$$\frac{400}{10^n} < 10 \text{ (single digit)}$$

$$\boxed{n=2} \quad \frac{400}{10^2} = \frac{400}{100} = 4.$$

$$e) y' = \frac{5}{x} \quad (1) \quad x=1, \quad n=5, \quad y=0. \quad (2)$$

$$y-0 = 5(x-1) \quad (3)$$

$$\boxed{y = 5x - 5}$$

ECT = error carried thru.

Question 12.

$$a) i) y = x^2 e^{-x} \quad \checkmark \text{ Product rule}$$

$$y' = 2x \cdot e^{-x} - x^2 \cdot e^{-x}$$

$$= x e^{-x} (2 - x)$$

$$d) i) 5^{3x} = 0.04$$

$$\log_5 0.04 = 3x$$

$$x = \frac{\log_5 0.04}{3}$$

$$(ii) y = \frac{x^2}{3x+1}$$

$$y' = \frac{(3x+1)(2x) - x^2(3)}{(3x+1)^2} \quad \checkmark$$

$$= \frac{6x^2 + 2x - 3x^2}{(3x+1)^2}$$

$$= \frac{3x^2 + 2x}{(3x+1)^2} \quad \checkmark$$

$$= \frac{\log_5 5^{-2}}{3}$$

$$= \frac{-2 \cdot \log_5 5}{3}$$

$$= \frac{-2}{3} \quad \checkmark$$

$$b) \int \frac{2x^2}{2x^3-3} dx$$

$$= \frac{1}{3} \int \frac{6x^2}{2x^3-3} dx \quad \checkmark$$

$$= \frac{1}{3} \ln(2x^3-3) + C \quad \checkmark$$

$$(ii) \log_2 x(x-3) = 2.$$

$$x(x-3) = 2^2$$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$$x = 4, -1$$

$$c) i) \alpha + \beta = \frac{4}{3} \quad \checkmark$$

$$ii) \alpha \beta = -\frac{1}{3} \quad \checkmark$$

$$iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha \beta}$$

$$= \frac{\frac{4}{3}}{-\frac{1}{3}}$$

$$= -4 \quad \checkmark$$

$$\log_2 \left(\frac{x}{x-3} \right) = 2 \quad \checkmark$$

$$\frac{x}{x-3} = 2^2$$

$$x = 4x - 12$$

$$(iv) \alpha \beta^3 + \beta \alpha^3 = \alpha \beta (\beta^2 + \alpha^2)$$

$$= \alpha \beta [(\alpha + \beta)^2 - 2\alpha \beta] \quad \checkmark$$

$$= -\frac{1}{3} \left[\left(\frac{4}{3} \right)^2 - 2 \left(-\frac{1}{3} \right) \right]$$

$$= -\frac{16}{27} - \frac{2}{9} = -\frac{22}{27} \quad \checkmark$$

$$3x = 12$$

$$x = 4 \quad \checkmark$$

Question 13.

a) i) $T_8: a + 2d = 23$

$T_{10}: a + 9d = 72$

$7d = 49$

$d = 7$

$\therefore a = 9$

(2)

ii) $S_{18} = \frac{18}{2} [2 \times 9 + (17 \times 7)]$
 $= 1233$

(2)

b) $a = 6$ $r = 3$.

$S_n = 1594320$

$\frac{6(3^n - 1)}{3 - 1} = 1594320$

$3^n - 1 = \frac{1594320}{3}$

$3^n = 531441$

$\ln \cdot 3^n = \ln 531441$

$n = \frac{\ln 531441}{\ln 3}$

$n = 12$

(2)

c) $f'(x) = 15(5x-1)^2$

$f(x) = \frac{15(5x-1)^3}{5 \times 3} + C$

$= (5x-1)^3 + C$

$f(0) = 10, 10 = (-1)^3 + C$

$10 = -1 + C$

$C = 11$

$f(x) = (5x-1)^3 + 11$

(2)

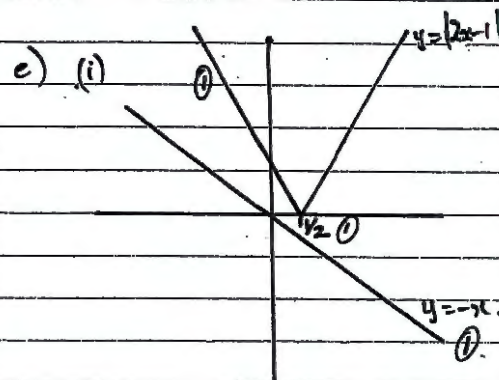
$P(x, y)$ $A(3, 1)$ $B(1, 3)$

d) $PA^2 = PB^2$

(i) $(x-3)^2 + (y-1)^2 = (x-1)^2 + (y-3)^2$
 $x^2 - 6x + 9 + y^2 - 2y + 1 = x^2 - 2x + 1 + y^2 - 6y + 9$
 $-6x - 2y + 10 = -2x - 6y + 10$
 $-4x + 4y = 0$
 $y = x$

(2)

(ii) Straight line with gradient +1, through (0,0) (1)



(3)

(ii) $|2x+1| = -x$
 $|2x+1| + x = 0$

(1)

No solutions since the graphs do not intersect.

Question 14

(a) (i) $x^2 - 4x = 2x - 5$ ✓
 $x^2 - 6x + 5 = 0$
 $(x - 5)(x - 1) = 0$ ✓ 2
 $x = 1, 5$ ✓

ii) Area = $\int_1^5 [(2x - 5) - (x^2 - 4x)] dx$
 $= \int_1^5 (-x^2 + 6x - 5) dx$ ✓
 $= \left[-\frac{x^3}{3} + 3x^2 - 5x \right]_1^5$ ✓
 $= \left(-\frac{125}{3} + 75 - 25 \right) - \left(-\frac{1}{3} + 3 - 5 \right)$
 $= 8\frac{1}{3} + 2\frac{1}{3}$ ✓ 3.
 $= \frac{32}{3} / 103 \text{ units}^2$ ✓

b) $\log_7 x^2 = 3$

$x^2 = 7^3$ ✓ 2
 $x = \sqrt{7^3}$ ✓
 $x = \pm \sqrt{343}$

c) $f(x) = x^3 + 6x^2 + 9x + 4$ $-4 \leq x \leq 1$

(i) $f'(x) = 3x^2 + 12x + 9$

SP $f'(x) = 0$ $x^2 + 4x + 3 = 0$ 3.
 $(x + 3)(x + 1) = 0$
 $x = -1, -3$

$f''(x) = 6x + 12$

$x = -1, f''(x) = -6 + 12 > 0 \therefore \text{min.}$

$x = -3, f''(x) = -18 + 12 < 0 \therefore \text{max.}$

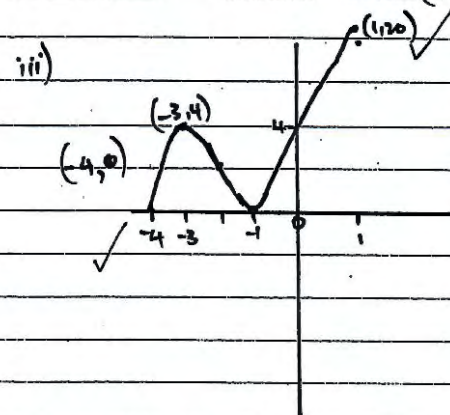
$(-1, 0)$ min. ✓

$(-3, 4)$ max. ✓

ii) Possible P.O.I. $f''(x) = 0$
 $6x + 12 = 0$
 $x = -2$

x	-3	-2	-1
$f''(x)$	-	0	+

change in concavity ✓
 $\therefore$ P.O.I. $(-2, 2)$



(iv) global max in this domain $y = 20$. ✓

15

Question 15.

a) $a = 20$
 $r = 0.8$ ✓

$$S_{\theta} = \frac{20}{1-0.8}$$

$$= 100 \text{ cm}$$

b) $x^2 - 8x - 16y + 48 = 0$

$$x^2 - 8x + 16 = 16y + 16 - 48$$

$$(x-4)^2 = 16y - 32$$

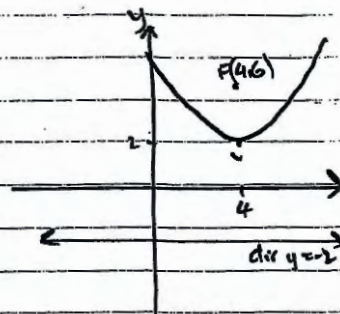
$$(x-4)^2 = 16(y-2)$$
 ✓

∴ Vertex (4, 2)

$$a = 4$$

Focus (4, 6)

Directrix $y = -2$ ✓



no mark if one is wrong.

c) LHS = $\frac{\tan \theta \cdot \sec \theta}{1 + \tan^2 \theta}$

$$= \frac{\tan \theta \cdot \sec \theta}{\sec^2 \theta}$$
 ✓

$$= \frac{\tan \theta}{\sec \theta}$$

$$= \frac{\sin \theta}{\cos \theta} \times \cos \theta$$
 ✓

$$= \sin \theta$$

$$= \text{RHS}$$

$$\text{RHS} = \sin \theta$$

(d) (i) $y = x \cdot e^x$
 $y' = x(e^x) + e^x(1)$
 $= xe^x + e^x$ ✓

(ii) $\int_0^2 \frac{xe^x}{2} dx$
 $= \frac{1}{2} [xe^x - e^x]_0^2$ ✓
 $= \frac{1}{2} [2e^2 - e^2 - (0 - 1)]$ ✓
 $= \frac{1}{2} (e^2 + 1)$ ✓

one mark deducted for incorrect coefficient

c) $\int_0^2 \frac{1}{x+1} dx$ 5 for values Trapezoidal Rule.

(i)

x	y	k	ky
0	1	1	1
$\frac{1}{2}$	$\frac{2}{3}$	2	$\frac{4}{3}$
1	$\frac{1}{2}$	2	1
$1\frac{1}{2}$	$\frac{2}{5}$	2	$\frac{4}{5}$
2	$\frac{1}{3}$	1	$\frac{1}{3}$

$$A \approx \frac{1}{2} \left(\frac{67}{15} \right)$$

$$= \frac{67}{60}$$
 ✓

(ii) $A = \int_0^2 \frac{1}{x+1} dx$
 $= [\ln(x+1)]_0^2$

$$\text{Diff} = \frac{67}{60} - \ln 3$$

$$= 0.018 \text{ (3 dp)}$$
 ✓

$$= \ln 3 - \ln 1$$
 ✓

$$= (\ln 3) \text{ units}^2$$
 ✓

Question 16.

a) $y = -16x^2 + 160x - 256$

$y' = -32x + 160$

SP when $y' = 0$ $160 = 32x$

$x = 5$

$y'' = -32 < 0 \therefore \text{Max.}$ $x = 5, y = 144$

Max y value is 144.

b) (i) $x + z + 6 = 16$

$x + z = 10$

$z = 10 - x$

(ii) $z^2 = x^2 + 36 - 12x \cos Z$

~~$z^2 = (10-x)^2 + 36 - 12x \cos Z$~~
 ~~$z^2 = 100 - 20x + x^2 + 36 - 12x \cos Z$~~

(iii) $(10-x)^2 = x^2 + 36 - 12x \cos Z$

$100 - 20x + x^2 = x^2 + 36 - 12x \cos Z$

$64 - 20x = -12x \cos Z$

$12x \cos Z = 20x - 64$

$\cos Z = \frac{20x - 64}{12x}$

$\cos Z = \frac{5x - 16}{3x}$

iv) $A = \frac{1}{2} \times 6x \cdot \sin Z$

$A = 3x \cdot \sin Z$

$\therefore A^2 = 9x^2 \cdot \sin^2 Z$

v) $A^2 = 9x^2 (1 - \cos^2 Z)$

$= 9x^2 \left(1 - \left(\frac{5x-16}{3x}\right)^2\right)$

$= 9x^2 - (5x-16)^2$

$= 9x^2 - (25x^2 - 160x + 256)$

$A^2 = -16x^2 + 160x - 256$

vi) $(A^2) = -16x^2 + 160x - 256$

from part (a), max value of $A^2 = 144$

$\therefore$ Max Value of $A = 12$

(c) $(k+1)x^2 - 4kx + 4k-3$

Sum: $x + \beta = \frac{4k}{k+1}$

Prod

$x\beta = \frac{4k-3}{k+1}$

$1 + \beta = \frac{4k}{k+1}$

$\beta = \frac{4k-3}{k+1}$

$\beta = \frac{4k-3}{k+1} - 1$

$\therefore \frac{4k-3}{k+1} - 1 = \frac{4k-3}{k+1}$

$4k - (k+1) = 4k-3$

$4k - k - 1 = 4k-3$

$-k = -2$

$k = 2$

(d) $y = \log_e x$ Area = 1

$$\therefore x = e^y$$

$$\text{Area} = \int_0^b e^y dy = 1$$

$$[e^y]_0^b = 1$$

$$e^b - e^0 = 1$$

$$e^b - 1 = 1$$

$$e^b = 2$$

$$\ln e^b = \ln 2$$

$$b = \ln 2$$

(3)